

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project Title

GramWatch – A Geo-Verified Community Grievance Reporting Platform for Rural India

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the complete functional and non-functional requirements of the **GramWatch** system. The document is intended to serve as an official approval and reference document for initiating the project under the Software Engineering Lab course. It acts as a contract between stakeholders and developers, ensuring a common understanding of system behavior and constraints.

1.2 Scope of the System

GramWatch is a mobile-based social grievance reporting system designed specifically for rural communities in India. The system enables villagers to report real-world civic and infrastructure issues such as road damage, water supply problems, sanitation issues, electricity failures, and public health concerns by uploading geo-tagged images or videos.

Instead of traditional social media likes, GramWatch uses **community validation** within the same geographic region to confirm the authenticity of reported issues. Once an issue reaches a predefined validation threshold, it is automatically escalated to the appropriate Panchayat authority. The system digitally tracks the complete lifecycle of every issue, ensuring transparency, accountability, and public awareness.

2. Overall Description

2.1 Product Perspective

GramWatch is a standalone, cloud-supported application that integrates a mobile user interface, backend processing services, and a web-based authority dashboard. The system leverages GPS services, cloud databases, and notification systems to enable real-time reporting and monitoring of rural issues.

2.2 Product Functions

The major functions of the GramWatch system include:

- User registration and authentication
- Issue reporting with multimedia support
- Automatic geo-location tagging
- Community-based issue validation
- Automated escalation to authorities
- Real-time issue status tracking
- Authority-side issue management
- Public awareness and browsing of issues

2.3 User Classes and Characteristics

User Class	Description
Villagers	Primary users who report issues; may have low digital literacy
Validators	Users within the same geographic area who validate issues
Panchayat Authorities	Officials responsible for resolving issues
Public Viewers	Citizens and youth viewing issues for awareness

2.4 Operating Environment

- Android-based mobile application
- Web browser for authority dashboard
- Cloud-hosted backend services
- Internet connectivity (optimized for low bandwidth)

2.5 Design Constraints

- Limited smartphone proficiency among rural users
- Intermittent network availability
- Academic project timeline constraints
- Budget limitations for hosting and services

2.6 Assumptions and Dependencies

- Users possess GPS-enabled smartphones
- Panchayat authorities participate in the system
- Users have minimum internet access

3. Functional Requirements

FR1: User Registration and Authentication

- The system shall allow users to register using a mobile number or email ID.
- The system shall authenticate users before granting access.

FR2: Issue Reporting

- The system shall allow users to upload images or videos of issues.
- The system shall automatically capture GPS location data.
- The system shall require users to select an issue category.

FR3: Community Validation

- The system shall allow only users within a predefined geographic radius to validate an issue.
- The system shall prevent multiple validations by the same user.

FR4: Validation Threshold Handling

- The system shall define a minimum validation count for escalation.
- The system shall continuously track validation counts.

FR5: Escalation to Authorities

- The system shall automatically forward validated issues to Panchayat authorities.
- The system shall notify authorities digitally.

FR6: Issue Status Tracking

- The system shall display issue status updates to users.
- The system shall maintain timestamps for every stage of issue handling.

FR7: Authority Dashboard Functions

- The system shall allow authorities to view assigned issues.
- The system shall allow authorities to update issue status and remarks.

FR8: Public Awareness Module

- The system shall allow public users to browse issues.
- The system shall support filtering by location and category.

4. Non-Functional Requirements

4.1 Performance Requirements

- The system shall respond to user requests within 3 seconds under normal load.

4.2 Security Requirements

- The system shall ensure secure user authentication.
- The system shall protect sensitive user data and media content.

4.3 Usability Requirements

- The system interface shall be simple and icon-driven.
- The system shall be usable by low-literacy users.

4.4 Reliability Requirements

- The system shall prevent data loss during temporary network failures.

4.5 Scalability Requirements

- The system shall support scaling from village level to district and state levels.
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5. External Interface Requirements

5.1 User Interfaces

- Mobile UI for villagers and validators
- Web-based dashboard for authorities

5.2 Hardware Interfaces

- Smartphone camera
- GPS sensor

5.3 Software Interfaces

- Cloud database services
 - Notification services
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6. Future Enhancements

- AI-based issue categorization
 - Multilingual language support
 - Data analytics for governance planning
 - Integration with government grievance portals
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7. Conclusion

This SRS document defines the complete functional and non-functional scope of the GramWatch system. It establishes a strong foundation for project approval and structured development under the Software Engineering Lab.